Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | Powder-Based Coating Strategies for Porous and Compact Materials | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Mechanical Materials | | Mechanical Materials | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Metal Materials | Functional Materials | Surface Treatment | Coating/Method Technologies |
| Concept of Proposed Technology | The proposed technology is based on powder-based coating approaches for the tailored modification of open porous metal foams, other porous substrates, or compact materials. By applying metallic powders, the composition, surface topography, and microstructure can be precisely adjusted, often resulting in a significant increase in the effective surface area. Different coating strategies, ranging from wet to dry processes, are available, enabling the generation of a wide variety of coating morphologies and functional gradients. Where required, post-deposition thermal treatments are applied to ensure a strong metallurgical or functional bonding between the coating and the substrate. | | | |

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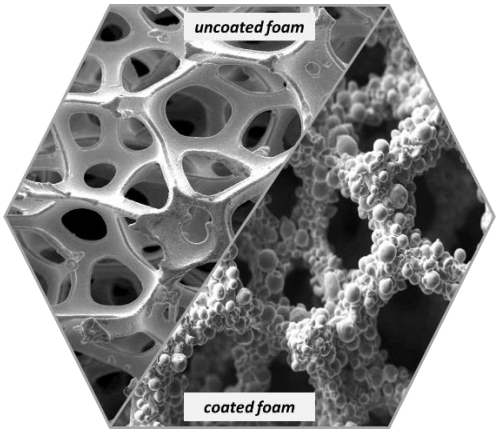
| |  |
| **Rationale for Joint R&D Project** | The necessity of this joint R&D project arises from established expertise in powder-based coating technologies and the capability to integrate different material classes, as demonstrated in prior research on electrode materials for alkaline water electrolysis and water treatment applications. Further advancement toward application-oriented and industrially relevant solutions requires close and structured collaboration with external partners. In particular, cooperation within a joint R&D framework enables the alignment of advanced coating concepts with specific industrial requirements and operating conditions. Such collaboration provides a platform for mutual exchange of expertise, joint development and validation of innovative material solutions, and supports sustainable, long-term international cooperation with Korean partners. |
| **Key Words** | 1. coating technology<br>2. surface modification<br>3. graded structures<br>4. open porous metal foams<br>5. powder metallurgy |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (3) | End of the Project | Level (5) |
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| R&D Objectives | The objective of this R&D project is to further develop existing powder-based coating technologies to enable the targeted integration of partner-specific material systems onto suitable substrates. The focus lies on adapting coating processes to new material combinations, such as functional and catalytic materials provided by project partners, while ensuring controlled composition and microstructural design. By employing tailored wet and dry coating routes, robust and application-oriented coating methodologies will be established to support the effective use of partner-developed materials in industrially relevant applications. | | | |
| R&D Target Specifications | The target performance specifications of this R&D project focus on the reliable application of partner-specific material systems through advanced powder-based coating technologies. Key measurable parameters include coating thickness in the range of approximately 10 - 200 µm, homogeneous or graded composition across the (porous) substrate, and controlled microstructural features such as homogeneity, composition and surface area. Additional performance targets include reproducibility of coating quality, scalability of the processes, and compatibility with functional materials such as catalytic or electrochemically active phases. | | | |
| R&D Plan | The R&D project is planned for a duration of 36 months and follows a structured and collaborative approach aimed at the development, characterization, and validation of advanced powder-based coating methodologies for partner-specific material systems.<br><br>**Phase 1: Definition of requirements and material selection (Months 1–3)**<br>In close collaboration with project partners, target applications, performance requirements, and relevant operating conditions will be defined. Partner-provided functional or catalytic materials will be analyzed, and suitable substrate materials will be selected as a basis for subsequent coating development.<br><br>**Phase 2: Evaluation and development of coating methodologies (Months 4–15)**<br>Different wet and dry coating approaches will be systematically evaluated to identify suitable methods for the selected material systems. This phase includes the development and optimization of coating parameters, formulation and adjustment of suspensions where required, and the application of coatings to substrates. If necessary, post-deposition thermal treatments such as debinding, annealing, and sintering will be performed to tailor microstructure, phase composition, and coating–substrate bonding. These treatments may be conducted under oxidizing (air), inert ($N_2$, Ar), or reducing ($H_2$) atmospheres, under high vacuum, and, where appropriate, using pressure-assisted techniques such as hot pressing or spark plasma sintering. In addition, substrate pre-treatments including pressing, rolling, bending, stamping, laser cutting, waterjet cutting, or electrical discharge machining (EDM) may be applied prior to coating. Where required, the coating powder itself can also be modified, for example through mixing or milling, to achieve the targeted coating performance. This phase focuses on establishing robust and reproducible coating routes.<br><br>**Phase 3: Parallel characterization and analysis (Months 1–36)**<br>Comprehensive characterization will be conducted in parallel throughout the project. This includes the analysis of raw partner materials, suspensions (if applicable), and coated samples. Key methods comprise microstructural analysis (SEM), compositional analysis (EDX, ICP, CNOHS gas analysis), X-ray diffraction (XRD), surface area and roughness (BET, Hg porositmetry, profilometer), powder analysis (particle size analysis, apparent and tap density, powder flow characterization) and wetting characterization using tensiometry. Depending on partner requirements, selected mechanical (tensile test, compression test, bending test, RFDA), thermal (DSC, DTA, TG, DIL), electrochemical and physical properties (electrical and thermal conductivity, pressure drop, magnetic properties) can also be evaluated to assess structure–property relationships. | | | |

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| | **Phase 4: Preparation of optimized samples for partner validation (Months 24–36)**<br>Based on the results obtained, coatings with the most promising performance will be reproduced and applied to substrates in sufficient quantity and quality. These samples will be provided to project partners for application-oriented testing and validation under relevant operating conditions.<br><br>**Key Milestones**<br>• **M1 (Month 12):** Successful coating of substrates using partner-provided material systems<br>• **M2 (Month 18):** Assessment of process robustness and scalability of selected coating methods<br>• **M3 (Month 30):** Production of optimized coated samples for testing and validation by project partners<br><br>The proposed R&D plan represents an initial framework and will be flexibly adapted in close coordination with the project partner to reflect specific requirements, priorities, and application needs. |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | Fraunhofer Institute for Manufacturing Technology and Advanced Materials (IFAM) has long-standing expertise in powder metallurgy and advanced coating technologies for functional materials. For more than two decades, the institute has been actively engaged in the development and application of coating processes for open-porous substrates, including metallic foams and other porous structures. These activities encompass the tailored deposition of metallic and functional powders, the control of microstructure and porosity, and the integration of coatings into application-oriented components. Fraunhofer IFAM contributes extensive know-how in process development, materials characterization, and the optimization of coating–substrate interactions, providing a strong foundation for collaborative R&D and industrial implementation. |
| **Expected R&R of Korean SME Partner** | The Korean SME is expected to play a key role in the joint R&D project by contributing its expertise in open porous structures, catalysis, water treatment, water electrolysis or similar. The partner will provide application-specific functional and catalytic materials and define relevant performance requirements and operating conditions. In parallel, the SME may further develop or newly design its material systems during the project in order to optimize functional performance. In close collaboration with Fraunhofer IFAM, these materials will be adapted for deposition onto high-surface-area substrates to enhance overall system efficiency. The Korean SME will be actively involved in application-oriented testing and validation under relevant conditions, provide technical feedback for iterative optimization, and contribute resources in terms of materials, technical know-how, and personnel. Continuous collaboration and knowledge exchange will ensure alignment of the developed coating solutions with industrial needs. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | The joint R&D project is expected to enable the industrial partner to obtain either an improved or newly developed material system that demonstrates enhanced performance when applied to high-surface-area substrates, or a clear understanding of the achievable performance of coated porous substrates. Through the collaboration, coating processes will be adapted and optimized for the partner's materials, providing validated knowledge on coating behavior, structure–property relationships, and process limitations. These outcomes will support informed decisions on further material development, process optimization, and potential commercialization, while reducing development risks and strengthening the partner's technological competitiveness. |

## 6. References

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |

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